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Graduation Project Report

Udjat Track

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Chapter 1 Introduction

1.1 Background and Motivation

Road traffic accidents constitute one of the most critical public safety challenges in Egypt. National injury surveillance data reported by the Egyptian Ministry of Health in collaboration with the World Health Organization show that road traffic injuries increased from 11.52% of total injuries in 2001 to 12.87% in 2009, reflecting a persistent deterioration in road safety conditions. Heavy transport trucks play a dominant role in this context, as more than 96% of goods in Egypt are transported by trucks, making their safety impact particularly significant. Due to their large size, excessive weight, and extended operating hours, truck-related accidents are often more severe than those involving passenger vehicles, and quantitative analysis on the Cairo–Suez Highway confirms that trucks have a statistically significant correlation with total road traffic accidents, with their contribution estimated at 0.92 times that of passenger cars.

Human factors have been identified as the primary contributors to truck-related crashes in Egypt. A large-scale study involving 615 Egyptian truck drivers found that fatigue, lack of sleep, prolonged continuous driving, obesity, and drug use are the most influential factors affecting crash involvement. The same study reports that drivers who slept less than 6 hours accounted for 59% of accident-involved drivers, while 41% of drivers involved in accidents drove more than 15 hours per day.

Furthermore, truck accidents are associated with high-severity outcomes. Survey-based accident data revealed that 13% of truck accidents resulted in deaths, while 19% resulted in injuries, and 49% caused property damage. In fatal truck accidents, 51% involved the death of at least one person, and 36% involved the death of two or more individuals, emphasizing the catastrophic nature of these incidents.

Despite these alarming figures, most heavy transport vehicles operating on Egyptian roads still lack intelligent driver monitoring, fatigue detection, and real-time accident response systems. Existing safety solutions are largely limited to basic GPS tracking, while advanced driver assistance technologies remain financially inaccessible for most local fleet operators. This gap between the high risk of truck-related accidents and the limited availability of effective safety technologies motivates the development of an intelligent, affordable, and locally adaptable safety system for heavy transport trucks.

1.2 Problem Statement

Heavy transport truck accidents in Egypt continue to occur at high rates due to a combination of human, technical, and systemic factors. Empirical studies confirm that driver fatigue, excessive continuous driving, distraction, and aggressive driving behavior are statistically significant predictors of accident occurrence among truck drivers.

Current safety practices rely heavily on post-accident reporting rather than real-time prevention. Most trucks lack integrated systems capable of detecting driver drowsiness, monitoring unsafe driving durations, or automatically reporting crashes when drivers are unconscious or unable to call for help. Research indicates that more than 30% of fatal truck crashes are fatigue-related, according to findings referenced by the National Transportation Safety Board.

Fleet operators also face limited visibility into driver behavior and vehicle safety conditions during long-haul trips. As a result, dangerous conditions such as micro-sleep, distraction, or prolonged driving often go unnoticed until a severe accident occurs. This absence of proactive monitoring creates a critical gap between existing road safety challenges and effective accident prevention mechanisms, leading to unnecessary loss of life, cargo damage, and economic disruption.

1.3 Objectives

In response to the challenges mentioned, the objectives of this project focus on improving safety and operational efficiency in heavy transport truck operations. The system aims to reduce accidents caused by driver fatigue, sickness, and unsafe driving behavior by continuously monitoring drivers during active trips and issuing timely alerts when abnormal conditions are detected. To address the lack of structured trip oversight and excessive continuous driving, the system supports comprehensive trip management by enabling drivers to create, track, and end trips while automatically logging into driving duration, break periods, and trip history.

In addition, to overcome limited fleet visibility and delayed intervention, the system provides fleet managers with centralized tools to manage driver accounts, monitor trip progress, and review safety alerts in real time. The integration of map-based services also addresses navigation challenges faced by drivers by displaying essential nearby facilities such as roads, hospitals, gas stations, and maintenance centers. Through these objectives, the project seeks to bridge the gap between current road safety challenges and proactive accident prevention by improving coordination between drivers, fleet managers, emergency services, and maintenance centers, ultimately contributing to safer and more efficient heavy transport operations.

1.4 Project Scope and Limitations

1.4.1 Project Scope

Udjat Track is a comprehensive, AI-powered safety and assistance platform designed specifically for the heavy transport sector in Egypt. The project scope encompasses the following core areas:

- **AI-Powered Driver Monitoring:** Utilising on-device computer vision to detect signs of **drowsiness, head nodding, and prolonged eye closure**, triggering immediate in-app alarms.
- **IoT Assistance:** Integrating ultrasonic sensors for **blind-spot and forward-collision warnings**, which are features often absent in older Egyptian truck models.
- **Emergency Response Automation:** Implementing **accident detection** via internal smartphone IMU sensors and an **automatic SOS system** that notifies emergency services and fleet managers if a driver is unresponsive.
- **Fleet Oversight Platform:** Developing a **companion web app** for remote real-time monitoring of vehicle locations, safety status, and incident logs.
- **Compliance and Analytical Reporting:** Generating automated **trip and safety log reports** in PDF format for insurance, compliance, and post-incident review.

1.4.2 Project Limitations

During the ongoing development of the system, several anticipated technical and environmental limitations are being considered and gradually addressed:

Hardware and Processing Constraints: Real-time video processing requires substantial computational resources, including RAM and CPU capacity. To ensure feasibility on embedded hardware, the design approach includes techniques such as resizing input frames, adopting adaptive frame rates.

Connectivity and Network Dependency: Heavy transport routes frequently traverse areas with unreliable cellular coverage. Since the system depends on the mobile network for cloud

synchronization, mechanisms such as local data buffering and network failover strategies are being incorporated to minimize data loss during connectivity interruptions.

Environmental Factors: The in-vehicle hardware is expected to operate under challenging conditions, including strong vibrations, high temperatures, and humidity. Accordingly, the system design includes plans for protective and durable casings for sensors and electronic components to enhance reliability in real-world deployment.

External Data Dependency: The “Nearby Help Map” functionality relies on third-party mapping services such as Google Maps or OpenStreetMap. As a result, its performance and availability remain dependent on the uptime and accessibility of these external APIs.

1.4.3 Initial Project Plan

The project followed a **Waterfall methodology**, chosen for its structured, linear approach which suits safety-critical systems requiring hardware integration and stable requirements. The plan consisted of five sequential phases:

1. **Requirements Analysis:** Eliciting functional and non-functional requirements through system analysis.
2. **System Design:** Designing the architecture, UI/UX, and UML diagrams (e.g., Use Case, Class).
3. **Implementation:** Developing the mobile app, web dashboard, and integrating IoT hardware.
4. **Testing:** Validating functionality, usability, and safety protocols.
5. **Deployment and Evaluation:** Final system deployment and performance assessment.

1.4.4 Team Members and Responsibilities

The project was developed by a multidisciplinary team within the **Information Systems Department** at FCAI-CU.

<i>Team Member</i>	<i>Main Responsibilities</i>
Maya Ahmed	AI & ML, Mobile Application Development.
Nadra Mahmoud	AI & ML, Mobile Application Development.
Duaa Abd-Elati	Back-End Development
Hana Hany	Front-End Development
Tasneem Gomaa	UI/UX Design, Front-End Development.
Mariam Matrawy	Back-End Development

1.5 Project Methodology

1.5.1 Selected Methodology

The Waterfall methodology follows a linear and sequential approach, where each phase must be completed before moving to the next. This methodology was chosen for the following reasons:

- The project requirements are clearly defined and stable, which makes a sequential development approach suitable.
- Udjat Track has a fixed scope and deadline, making Waterfall easier to manage and document.
- Each phase (analysis, design, implementation, testing, and deployment) can be reviewed and approved before proceeding, reducing the risk of major changes later.
- The system involves hardware integration and safety-critical features, which benefit from careful upfront planning and structured execution as offered by the waterfall.

1.5.2 Waterfall Phases Applied to the Project

- **Requirements Analysis:**
Gathering functional and non-functional requirements through user research and system analysis.
- **System Design:**
Creating system architecture, information architecture, wireframes, and UI/UX designs.
- **Implementation:**
Developing the mobile application, web dashboard, and integrating hardware components.
- **Testing:**
Conducting functional testing, usability testing, and safety validation.
- **Deployment and Evaluation:**
Deploying the final system and evaluating its performance and usability.

1.6 Project Outline

- **Chapter 1:** presents the background and motivation for the project, the problem statement regarding heavy transport safety in Egypt, the project objectives, scope, limitations, and the Waterfall methodology adopted for development.
- **Chapter 2:** presents the literature review of heavy transport safety challenges, the technological gap in existing truck fleets, a comparison between Udjat Track and OEM-integrated ADAS, and an analysis of existing commercial solutions and driver risk factors.
- **Chapter 3:** presents the analysis of the Udjat Track system, including its core problem, stakeholder roles, functional and non-functional requirements, detailed system scenarios, and system modeling through various UML diagrams.
- **Chapter 6:** presents the conclusions regarding the potential of the Udjat Track system as a safety intervention and outlines future work such as advanced AI adaptation, insurance integration, and intelligent break management.

Chapter 2 Market and Literature Survey

2.1 Justification for Focusing on Heavy Transport Vehicles

Heavy transport vehicles, particularly trucks, represent a critical focus area in road safety research due to their disproportionate involvement in severe traffic accidents. A study conducted in 2020 on Egyptian road traffic conditions indicates that heavy vehicles contribute significantly to serious crashes as a result of their large mass, heavy payloads, and longer stopping distances compared to light vehicles. Expert analyses estimate that trucks are involved in approximately 20–40% of road collisions in Egypt, many of which result in severe injuries or fatalities.

Evidence focusing specifically on truck-related accidents further highlights the severity of this vehicle category. Data derived from a detailed study of Egyptian truck drivers shows that while 49% of truck crashes result in vehicle damage only, a combined 38% lead to injuries or deaths, and 13% result in fatalities alone. Compared to general road accident statistics, these figures indicate that truck accidents tend to produce more severe outcomes, reinforcing the need for targeted safety interventions rather than generalized traffic safety measures.

Statistical analysis of accident patterns on major freight corridors further supports this focus. A study conducted on the Cairo–Suez Highway reveals a strong positive correlation between truck-involved accidents and the total number of road traffic accidents (Pearson correlation = 0.891, significant at the 0.01 level). This finding indicates that increases in truck-related accidents substantially influence overall accident occurrence, identifying trucks as a key predictive factor in traffic safety outcomes on this critical highway.

Taken together, the high level of truck involvement in severe accidents, their statistically significant contribution to overall accident occurrence, and their essential role in freight transportation provide a clear and data-driven justification for selecting heavy transport vehicles as the primary focus of this project. Interventions targeting truck safety, therefore, have the potential to produce meaningful improvements in overall road safety conditions in Egypt.

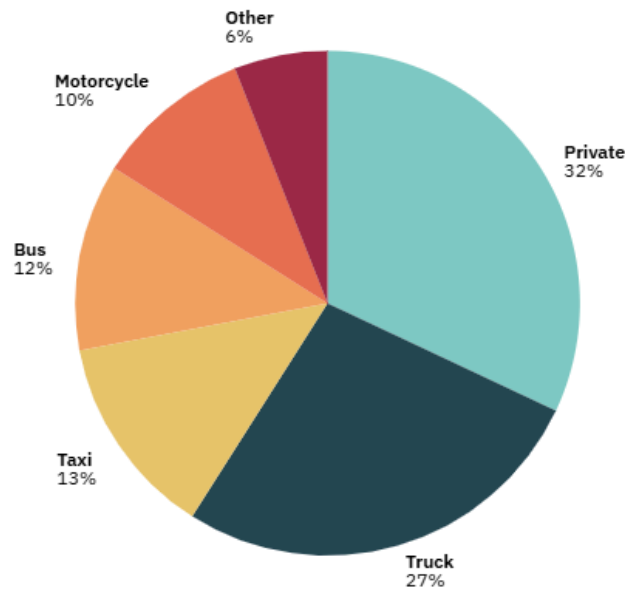


Figure 1: Percentage of crashes by vehicle type in Egypt

2.1.1 Technology Gap in Existing Heavy Truck Fleets

Industry evidence suggests that, despite the rapid expansion of the Egyptian commercial vehicle market, the majority of heavy trucks currently in operation are not equipped with modern active safety technologies such as Advanced Driver Assistance Systems (ADAS). The Egypt Trucks Market Report emphasizes strong growth in market value from USD 0.53 billion in 2024 to an estimated USD 1.51 billion by 2030, driven by infrastructure development and rising logistics demand. In parallel, the Egypt Fleet Management Industry is valued at approximately USD 210 million, reflecting the increasing digitalization of fleets driven by e-commerce growth, urbanization, and demand for cost-effective logistics operations, particularly in Cairo and Alexandria, the country's primary transport hubs. However, the report

does not indicate that ADAS or other advanced safety systems are widely integrated into standard heavy-truck configurations, implying that fleet modernization has prioritized capacity and transport efficiency rather than embedded vehicle safety intelligence.

This inference is reinforced by analyses of the Egyptian fleet management sector, which show that technological adoption has primarily focused on basic telematics and operational monitoring rather than proactive safety automation. The Egypt Fleet Management Industry Report highlights increasing use of GPS tracking and real-time monitoring, further strengthened by 2023 regulations mandating certified tracking devices for commercial vehicles, as well as a market structure in which the services segment (fleet tracking, maintenance, and fuel management) dominates, demonstrating a primary focus on operational efficiency rather than driver safety automation. Similarly, the Truck Auxiliary Braking Device Market Report indicates that even fundamental safety enhancements remain unevenly adopted and are largely limited to braking performance rather than comprehensive electronic safety or driver-support systems.

Collectively, these findings indicate that while digital monitoring technologies are becoming more common in Egyptian fleets, with an estimated 30% of fleets expected to adopt IoT and telematics solutions, these systems primarily support route optimization, fuel efficiency, and compliance rather than real-time driver safety assistance. Most heavy trucks, particularly in logistics, construction, and industrial sectors, therefore continue to operate without built-in ADAS or comparable advanced safety functionalities. This persistent technological gap underscores the necessity of affordable retrofit solutions such as the Udjat Track system that can provide real-time driver support, hazard detection, and proactive collision prevention for vehicles that lack integrated modern safety technology.

2.1.2 Comparison Between Udjat Track and OEM-Integrated ADAS

Advanced Driver Assistance Systems (ADAS) are widely recognized for improving road safety through technologies such as automatic emergency braking, adaptive cruise control, lane departure warnings, and blind-spot detection. The commercial vehicle ADAS market is projected to grow at approximately 13.6% CAGR between 2024 and 2029, reflecting increasing interest in automation and safety technologies for heavy vehicles.

However, despite this projected growth, several barriers limit the widespread adoption of ADAS in existing heavy truck fleets. A major constraint is the high cost of ADAS hardware, installation, and calibration, with full systems costing around USD 3,000 per vehicle. This makes adoption particularly difficult in price-sensitive and emerging markets, where fleet operators tend to prioritize basic operational needs over advanced safety technologies.

Beyond cost, technical and acceptance barriers further slow ADAS penetration. Many heavy trucks especially older models lack the necessary electronic architecture to support factory-level ADAS, making retrofitting complex and expensive. Additionally, some drivers and small fleet operators express concerns about system reliability, maintenance costs, and unclear direct benefits, which further discourages adoption. As a result, ADAS installation rates in commercial heavy vehicles remain relatively low (often reported below 20%), leaving a large proportion of fleets without advanced safety support.

These limitations are particularly relevant in contexts such as Egypt, where many heavy trucks are older models and fleet modernization has focused mainly on GPS tracking and basic telematics rather than advanced safety automation.

In contrast, Udjat Track is designed as a low-cost, retrofit-friendly alternative for heavy trucks that lack OEM ADAS. Using modular edge-based sensors such as IMU for crash detection, ultrasonic sensors for blind-spot and proximity warnings, and AI-based driver fatigue monitoring the system provides practical, real-time safety support without requiring deep vehicle integration or expensive proprietary hardware.

Unlike traditional ADAS, which primarily focuses on external vehicle environment monitoring and automated control, Udjat Track emphasizes driver state awareness, proximity hazard detection, and automated emergency response, directly targeting fatigue- and distraction-related risks that are highly relevant to long-haul truck operations. This makes it a feasible and effective safety solution for fleets where full ADAS adoption remains impractical.

2.2 Literature Survey: Causes of Heavy-Truck Accidents and Risk Factors

2.2.1 Driver Fatigue and Driving Duration

In the Egyptian context, the study “Factors Affecting Accident Risks among Truck Drivers in Egypt” (ICTTE) demonstrates a strong relationship between accident involvement and fatigue-related factors, particularly insufficient sleeping hours, extended continuous driving, and excessive total driving hours. The findings indicate that drivers who sleep less than six hours exhibit substantially higher accident involvement rates compared to those who obtain adequate rest. Similarly, accident occurrence increases progressively as continuous driving duration exceeds four hours and as total daily driving time extends beyond ten hours. These results confirm that fatigue accumulation, rather than isolated driving events, plays a decisive role in elevating crash risk among truck drivers.

Further evidence from the ICTTE study demonstrates that accident occurrence among drivers sleeping less than six hours increases significantly with longer driving durations. For drivers with insufficient sleep, accident involvement rises from 13% when continuous driving is less than four hours to 34% when it exceeds eight hours. A similar pattern is observed for total driving hours, where accident occurrence increases from 9% for drivers operating less than five hours to 41% for those driving more than fifteen hours. These findings indicate that fatigue risk is cumulative and strongly influenced by the interaction between sleep deprivation and prolonged driving exposure.

Table 1: Percentage Distribution of Accident-Involved Truck Drivers According to Sleep Duration, Continuous Driving Time, and Total Driving Hours

Sleeping Hours	Less than 6 hours	7 - 8 hours.	More than 8 hours	
	59%	36%	5%	
Continuous Driving	Less than 4 hours	From 4 to 6 hours	From 7 to 8 hours	More than 8 hours
	13%	24%	29%	34%
Total Driving Hours	From 1 to 5 hours	From 6 to 10 hours	From 11 to 15 hours	More than 15 hours
	9%	21%	28%	41%

2.2.2 Aggressive and Risky Driving Behaviors

Aggressive and risky driving behaviors constitute another significant category of factors associated with accident involvement among truck drivers. The ICTTE study assessed such behaviors using frequency-based categories, not at all, sometimes, and often, to capture how frequently drivers engage in specific actions.

The results show that a considerable proportion of drivers reported sometimes or often engaging in aggressive behaviors. For example, 44.6% of drivers reported sometimes engaging in racing behavior, and 11.2% reported doing so often, while 44.2% reported not racing at all. Speeding behavior was reported sometimes by 42.9% of drivers and often by 16.4%, compared to 40.5% who indicated they did not speed.

Additional aggressive tendencies were also observed. Self-overconfidence was reported sometimes by 36.4% of drivers and often by 9.1%, while traffic violations were reported sometimes by 34.1% and often by 9.6%. Collectively, these findings indicate that aggressive driving behaviors are not isolated incidents but recurrent practices among a substantial subset of truck drivers, particularly when combined with fatigue and extended driving hours.

Table 2: Frequency Distribution of Aggressive Driving Behaviors among Accident-Involved Truck Drivers

Aggressive Driving	Variable	Not at all	Sometimes	Often
	Racing	44.2%	44.6%	11.2%
	Speeding	40.5%	42.9%	16.4%
	Self-Overconfidence	54.5%	36.4%	9.1%
	Drug use traffic offense	56.3%	34.1%	9.6%

2.3 Driver Survey Results: Safety Needs and Technology Acceptance

To complement the market analysis and literature survey, a small-scale driver survey was conducted involving 10 heavy-truck drivers operating in Egypt. The survey aimed to capture practical insights related to fatigue, safety risks, and acceptance of in-vehicle assistance technologies.

The results indicate that driver fatigue is common, with most participants reporting feeling fatigued while driving either sometimes or often. In addition, the majority of drivers expressed acceptance of in-cabin alerts for fatigue or loss of attention, indicating openness toward supportive safety systems. Audible alerts, either alone or combined with visual warnings, were preferred over visual-only alerts, suggesting the need for immediate and attention-grabbing feedback.

Drivers also reported frequent exposure to proximity and blind-spot risks, and most believed that proximity alerts would help reduce collision likelihood. In the event of an accident, respondents favored automated response mechanisms, such as automatic emergency notification and location sharing, over manual phone calls.

Overall, the survey findings support the inclusion of fatigue monitoring, proximity hazard detection, audible alert mechanisms, and automatic accident notification as core functionalities of the proposed system to match drivers' needs.

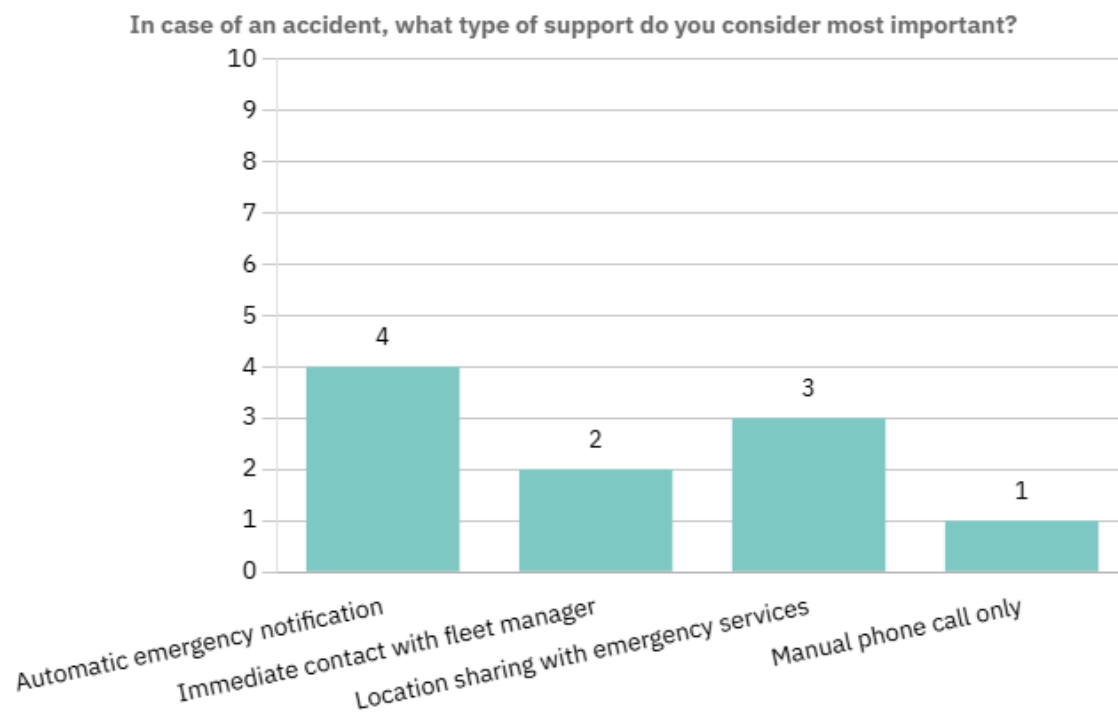


Figure 2: Driver Preferences for Accident Response and Support Mechanisms ($n = 10$)

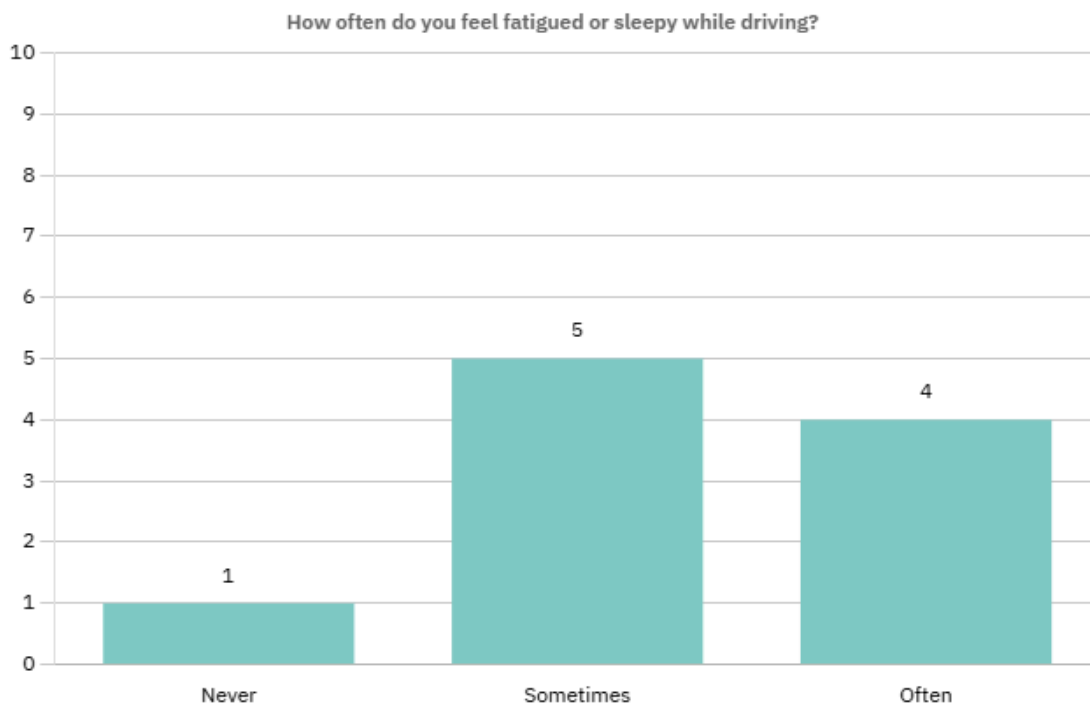


Figure 3: Frequency of Driver Fatigue While Driving ($n = 10$)

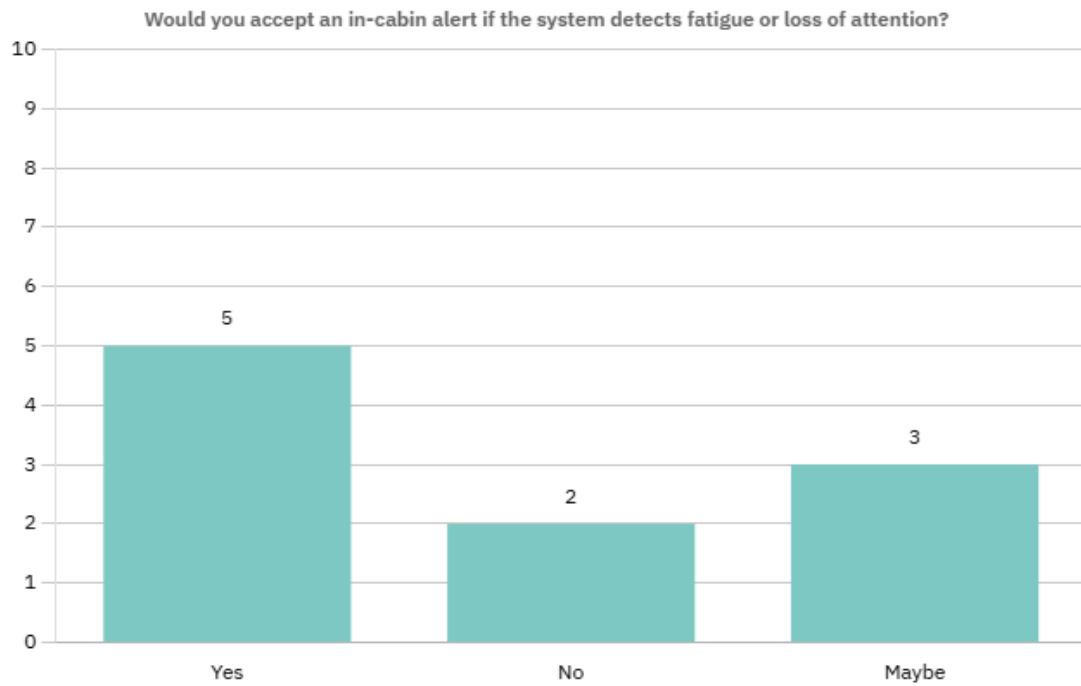


Figure 4: Driver Acceptance of In-Cabin Fatigue and Attention Alerts ($n = 10$)

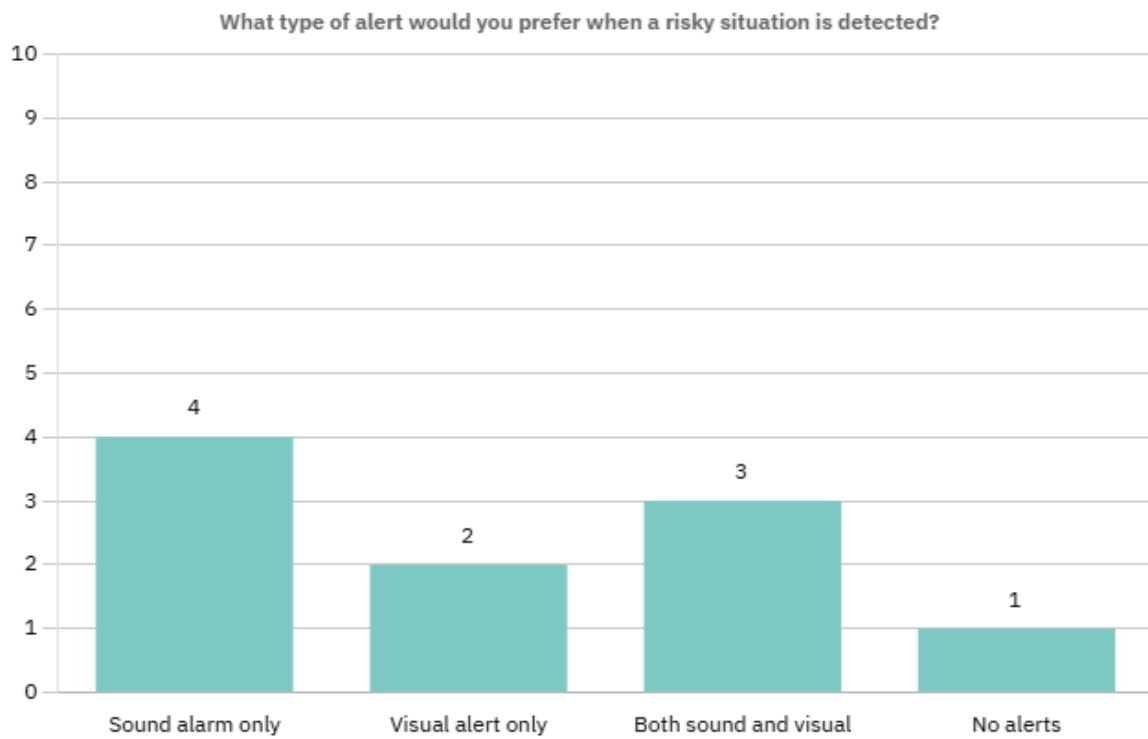


Figure 5: Preferred Alert Modalities During Risky Driving Situations ($n = 10$)

2.4 Existing Solutions and Market Landscape

The increasing focus on road safety and fleet efficiency has led to the development of several commercial driver monitoring and fleet management solutions. These systems primarily aim to reduce accident risk, improve regulatory compliance, and enhance operational efficiency. A review of existing solutions highlights their core capabilities, underlying design approaches, and limitations, particularly with respect to cost, deployment flexibility, and suitability for older heavy transport vehicles.

2.4.1 Techurity: Driver Monitoring System (DMS)

Techurity provides an artificial intelligence–based driver monitoring system that utilizes in-cabin cameras to detect driver fatigue, distraction, and inattentive behavior. The system generates real-time alerts and uploads video streams and behavioral analytics to cloud-based dashboards accessible by fleet managers. While the solution demonstrates strong capabilities in driver state monitoring and regulatory compliance, it relies on proprietary hardware and continuous cloud connectivity, which may increase deployment cost and limit scalability for small or medium-sized fleets.

2.4.2 FMS Tech: Fleet Management Solutions

FMS Tech offers a comprehensive fleet management platform that integrates GPS tracking, driver behavior analysis, fuel management, and vehicle diagnostics. The system emphasizes operational efficiency through route optimization, performance reporting, and centralized cloud dashboards. Although the platform provides extensive fleet-level insights, it is primarily designed for logistics and operational monitoring rather than real-time safety intervention, and it depends on specialized onboard devices and subscription-based services.

2.4.3 Softec: Altari Fleet Management System

Softec's Altari FMS is an enterprise-level fleet management solution targeting large-scale logistics and corporate fleets. The platform combines telematics data, predictive maintenance, and driver performance analytics to support compliance and operational decision-making. While Altari FMS is well-suited for high-volume commercial operations, its reliance on proprietary hardware, complex software modules, and enterprise-scale deployment models may limit its accessibility for smaller fleets or retrofitting older vehicles.

2.4.4 Udjat Track (Proposed System)

Udjat Track is proposed as a low-cost, sensor-based smart safety and monitoring system designed specifically for heavy transport vehicles. Unlike existing commercial solutions that rely heavily on proprietary hardware and cloud-centric processing, Udjat Track adopts an edge-processing approach using a Raspberry Pi and onboard sensors. The system detects risky conditions such as proximity hazards, driver fatigue indicators, and crash events locally, enabling faster response and reduced dependency on continuous network connectivity. Alerts and summarized data are transmitted to a backend system through a mobile application, allowing real-time awareness while maintaining affordability.

The system is intentionally designed to be retrofit-friendly, making it suitable for older trucks that lack built-in advanced driver assistance systems. By prioritizing safety-focused detection, local decision-making, and cost efficiency, Udjat Track addresses a gap in the current market between high-end enterprise fleet platforms and the practical needs of small to medium heavy-transport operators.

2.4.5 Functionality Comparison:

Table 3: Competitors' Functionalities Comparison Table

Feature / System Capability	Udjat Track (Proposed System)	Techurity DMS	FMS Tech	Softec Altari FMS
GPS Tracking	Yes (via mobile)	Yes	Yes	Yes
Trip Creation & Logs	Yes	Limited	Yes	Yes
Driver Drowsiness Detection	Via IMU behavior patterns	AI Camera-Based	No	No
Blind Spot / Proximity Detection	Ultrasonic Sensor	No	No	No
Crash Detection	IMU-based	Video-based	No	No
Driver Status Monitoring	Yes	Yes	Limited	Limited
Vehicle Motion Monitoring	Yes (IMU)	Camera Only	No	Basic
Real-Time Alerts to Mobile	Yes	Yes	Dashboard Only	Dashboard Only
Edge Processing (on-device)	Yes	Partial	No	No
Works on Old Trucks (Retrofit)	Designed for old vehicles	Needs installation	Needs CAN bus	Needs vehicle interface
Proprietary Hardware Required	No	Yes	Yes	Yes
Uses Open Components	Raspberry Pi & Sensors	Closed system	Closed system	Closed system

Installation Complexity	Simple	Medium	Complex	Complex
Maintenance Cost	Very Low	High	High	High
Cloud Dependency	Only for storage & dashboards	Heavy	Heavy	Heavy
Focus on Safety Events	Core Feature	Camera Events	Limited	Limited
FleetManagement Tools	Basic	Limited	Advanced	Advanced

Chapter 3 Udjat Track Analysis

3.1 Main Problem

Heavy transport vehicles pose a critical road safety challenge due to their disproportionate involvement in severe traffic accidents and the structural risks associated with their operation. The large mass and dimensions of trucks increase stopping distances and collision energy, leading to higher crash severity and greater harm to both truck drivers and other road users.

Statistical analyses indicate that truck traffic contributes significantly to overall accident occurrence and represents a strong predictor of accident frequency, with a greater impact than passenger vehicles. In addition, truck operations are commonly characterized by extended driving hours, limited rest opportunities, and high occupational pressure, which increase exposure to fatigue, distraction, and risky driving behaviors.

Despite these elevated risks, many heavy transport vehicles, particularly older fleets, lack integrated safety systems capable of monitoring driver condition, detecting proximity hazards, or providing automatic accident response. Existing solutions are often costly or oriented toward fleet management rather than real-time safety intervention.

Accordingly, the main problem addressed in this project is the lack of an affordable, real-time safety monitoring and intervention system tailored to heavy transport vehicles, capable of reducing accident risk through early detection, timely alerts, and automated response mechanisms.

3.2 Stakeholders and Responsibilities

3.2.1 Primary System Users

3.2.1.1 Driver

Stakeholder	Driver
Role	Primary end-user of the mobile application and in-vehicle IoT system
Responsibilities	• Start and end trips • View trip logs and trip progress • View maps and nearby facilities (hospitals, gas stations, maintenance centers) • Respond to real-time fatigue, drowsiness, and blind-spot alerts • View blind-spot warnings generated by the IoT sensor system • Report internal maintenance issues • Initiate SOS requests when required

3.2.1.2 Fleet Manager

Stakeholder	Fleet Manager
Role	Primary user of the web-based dashboard
Responsibilities	• Register, update, and delete driver accounts • Add, update, and delete vehicles • Assign drivers and vehicles to trips • Create and monitor trips • Track real-time driver and vehicle status • View system alerts and safety notifications • Monitor trip progress and driver behavior

3.2.2 Business and Commercial Stakeholders

3.2.2.1 Heavy Transport Companies

Stakeholder	Heavy Transport Companies
Role	System owners and operational beneficiaries
Objectives	• Reduce accident rates and vehicle downtime • Improve driver safety and compliance • Protect cargo and fleet assets • Enhance operational efficiency

3.2.2.2 Insurance Companies

Stakeholder	Insurance Companies
Role	Risk assessment and financial coverage entities
Objectives	• Analyze trip and safety logs • Assess fleet risk levels • Offer reduced premiums to compliant fleets

3.3 Functional requirements

3.3.1 Authentication and User Management

FR-01: Super Manager Authentication

The system shall allow the Super Manager to log in and manage the startup account.

FR-02: Fleet Manager Signup and Validation

The system shall allow fleet managers to sign up, select a subscription plan, and submit required validation criteria (e.g., business license, fleet size). Super Manager approval shall be required before login.

FR-03: Fleet Manager Login

The system shall allow validated fleet managers to securely log in with role-based access.

FR-04: Driver Account Creation and Assignment

Fleet Managers shall be able to register drivers, assign them to specific vehicles and trips, and manage their access.

FR-05: Driver Authentication

Drivers shall log in to the mobile application using credentials created by their fleet manager. The system shall enforce role-based access control

3.3.2 Trip Operations

FR-06: Trip Creation

Fleet managers shall be responsible for adding trips by specifying essential details such as the scheduled start and end time, starting location and destination, and by adding the assigned vehicle and a qualified driver to each trip.

FR-07: Trip Lifecycle Management

Drivers shall start, and end trips. Trips shall log start/end times, routes, duration, and break periods.

FR-08: Local Trip History Access

Drivers shall view trip history and logs locally.

FR-09: Trip Event Logging

The system shall record all operational and behavioral events during trips for real-time monitoring and future analysis.

3.3.3 Driver Monitoring and Fatigue Detection

FR-10: Facial Video Acquisition

Continuous capture of driver facial video via in-vehicle camera.

FR-11: Face Detection and Tracking

Real-time detection and tracking of driver faces.

FR-12: Driver State Classification

Classify drivers as Alert, Drowsy, High-Risk, or Unconscious based on combined biometric and temporal data.

3.3.4 Safety Alerts

FR-13: Real-Time Safety Alerts

Audio and visual alerts for unsafe driving conditions.

FR-14: Break Recommendation

Notifications when fatigue thresholds are exceeded.

3.3.5 Incident Reporting and Emergency Handling

FR-15: Manual Internal Issues Reporting

Drivers report internal vehicle issues. Reports forwarded to fleet managers in real-time.

FR-16: Manual SOS Alert

Drivers can send SOS manually if conscious. SOS includes real-time location and driver state sent to fleet managers.

FR-17: Automatic SOS Alert

Automatic SOS is sent in case the driver is unconscious, which includes real-time location and driver state sent to fleet managers, and fleet manager calls SOS.

FR-18: Crash Detection

Detect crash-level impacts.

FR-19: Driver Responsiveness Verification

The countdown timer verifies responsiveness after unconsciousness or crash detection.

FR-20: Emergency Stakeholder Notification

The fleet manager is notified in real time.

3.3.6 Proximity and Collision Warning

FR-21: Blind-Spot Detection

Side ultrasonic sensors detect nearby obstacles.

FR-22: Forward Distance Measurement

Detect obstacles in front using forward sensors.

FR-23: Collision Warning Delivery

Generate warnings in mobile apps when risk is detected.

3.3.7 Navigation and Location-Based Services

FR-24: Interactive Map Visualization

Drivers and fleet managers can view live maps, vehicle location, nearby hospitals, gas stations, and maintenance centers

FR-25: External Map API Integration

The system shall retrieve live map data, and location info from external APIs (e.g., Google Maps).

3.3.8 Fleet Management

FR-26: Real-Time Fleet Monitoring

Fleet managers view real-time vehicle location and driver state.

FR-27: Live Event Feed

Fleet managers receive alerts for accidents, fatigue, and SOS.

FR-28: Remote Trip Termination

Fleet managers can terminate trips remotely during emergencies.

FR-29: Predictive Maintenance

Monitor vehicle usage and notify managers when thresholds are reached.

FR-30: Report Generation

Generate safety, compliance, and trip reports.

3.3.9 System and Edge Operations

FR-31: Edge-Based AI Inference

Real-time processing on vehicle edge units.

FR-32: Sensor Data Acquisition and Transmission

Collect GPS, IMU, and ultrasonic sensor data at specified frequencies.

FR-33: Network Failover and Offline Data Storage

Maintain local data integrity during network outages; synchronize with cloud when connectivity is restored.

3.4 Non-Functional Requirements

3.4.1 Performance

NFR-01: Low-Latency Monitoring: Notifications delivered <500 ms.

3.4.2 Reliability and Availability

NFR-02: Mobile App Reliability: Run continuously as a foreground service.

NFR-03: Fault Tolerance: Local storage during outages

3.4.3 Security

NFR-04: Role-Based Access Control: Super Manager, Fleet Manager, Driver.

3.4.4 Scalability

NFR-06: Dynamic Cloud Resource Scaling: The system shall dynamically allocate cloud resources to ensure that the average API response time remains below 1 second, and does not exceed 3 seconds under peak load conditions.

3.4.5 Usability

NFR-07: Mobile UI: Distraction-free, high-contrast/night mode.

NFR-08: Dashboard UI: Intuitive fleet monitoring and alerts.

NFR-09: Alert Comprehensibility: Immediate understanding and actionability.

3.4.6 Accuracy and Robustness

NFR-10: Sensor/Edge Robustness: Vibration, temperature, humidity tolerance.

3.4.7 Maintainability

NFR-11: Modular Architecture: Independent upgrade/replacement of subsystems.

NFR-12: Logging and Diagnostics: Error and event logs accessible for troubleshooting.

3.4.8 Compliance

NFR-13: Regulatory Compliance: Automotive safety and transport regulations.

NFR-14: Communication Standards: Wi-Fi (IEEE 802.11) and cellular standards.

3.5 System Scenarios



Figure 6: AI-Based Driver Fatigue Detection and Accident Prevention Using Raspberry Pi Camera

Description:

This figure presents an AI-based driver fatigue detection system operating inside a long-haul cargo truck during night driving. A Raspberry Pi camera mounted on the dashboard continuously captures the driver's face and processes eye-closure and blink metrics locally using edge AI techniques. When fatigue indicators exceed safe thresholds, the system classifies the driver as drowsy and sends the status to a smartphone fixed on the dashboard. The mobile application displays “Drowsy Detected” and triggers an audio alert to warn the driver without requiring manual interaction. The driver then pulls over to rest, after which normal eye behavior is detected and the system status returns to “Safe,” demonstrating effective early intervention and accident prevention.

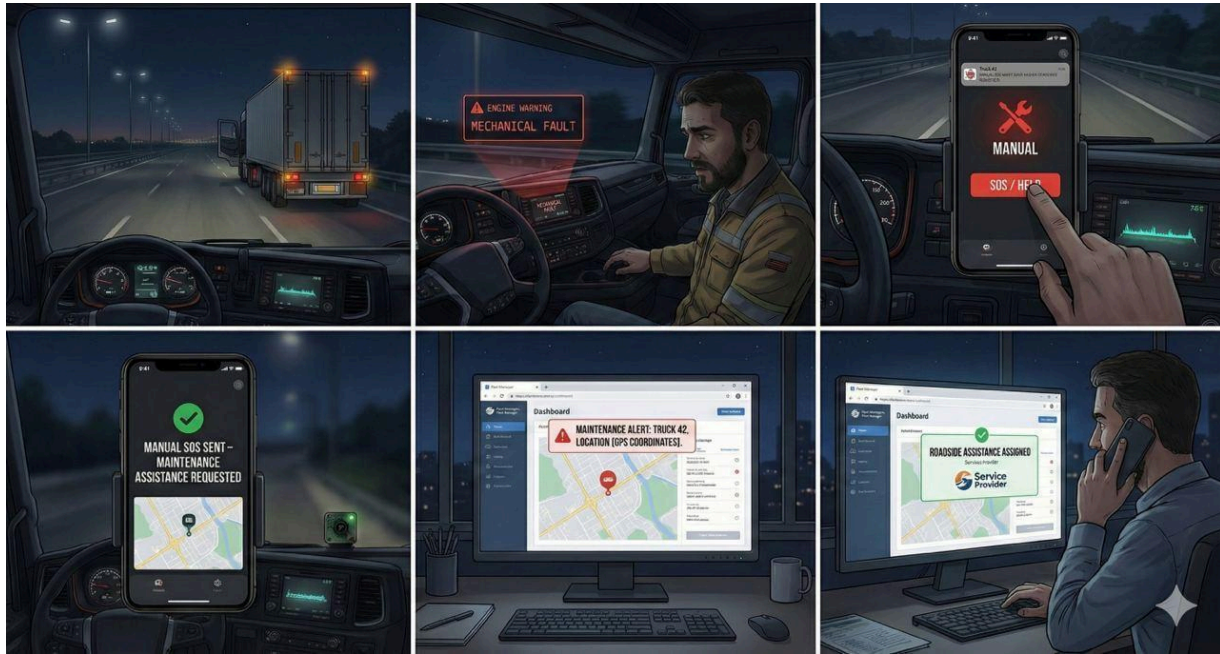


Figure 7: Manual SOS–Based Vehicle Breakdown and Maintenance Assistance Workflow

Description:

This figure illustrates the operation of the manual SOS feature for handling non-medical vehicle breakdowns during highway transport. When a heavy truck experiences a mechanical or maintenance issue, such as an engine warning or tire problem, the driver safely pulls over to the roadside and activates hazard lights. Using a smartphone securely mounted inside the cabin, the driver manually triggers the “SOS / Help” function within the mobile application. The system confirms that a maintenance assistance request has been sent. The alert is received on the fleet manager’s web dashboard, where the truck’s real-time GPS location is clearly displayed on a map interface. The fleet manager then assigns appropriate roadside assistance or maintenance support, enabling fast coordination and minimizing trip disruption through direct, non-emergency communication.



Figure 8: Blind-Spot Monitoring System for Heavy Trucks in Urban Traffic

Description:

This figure illustrates the operation of a blind-spot monitoring system for a heavy transport truck navigating busy urban streets in Cairo. As the truck prepares to change lanes, surrounding traffic includes nearby cars and motorcycles, one of which is positioned within the truck's blind spot and not visible through mirrors. Ultrasonic sensors mounted on the vehicle continuously measure the distance to nearby objects and detect the presence of the hidden motorcycle. The processed proximity data is sent to a smartphone securely mounted on the dashboard, where the mobile application displays a warning message, "Vehicle in blind spot," accompanied by an audible alert. The driver responds by stopping the lane change maneuver, allowing the motorcycle to pass safely. The system successfully prevents a potential collision, demonstrating how blind-spot detection enhances urban driving safety and reduces the risk of serious injuries and fatalities.

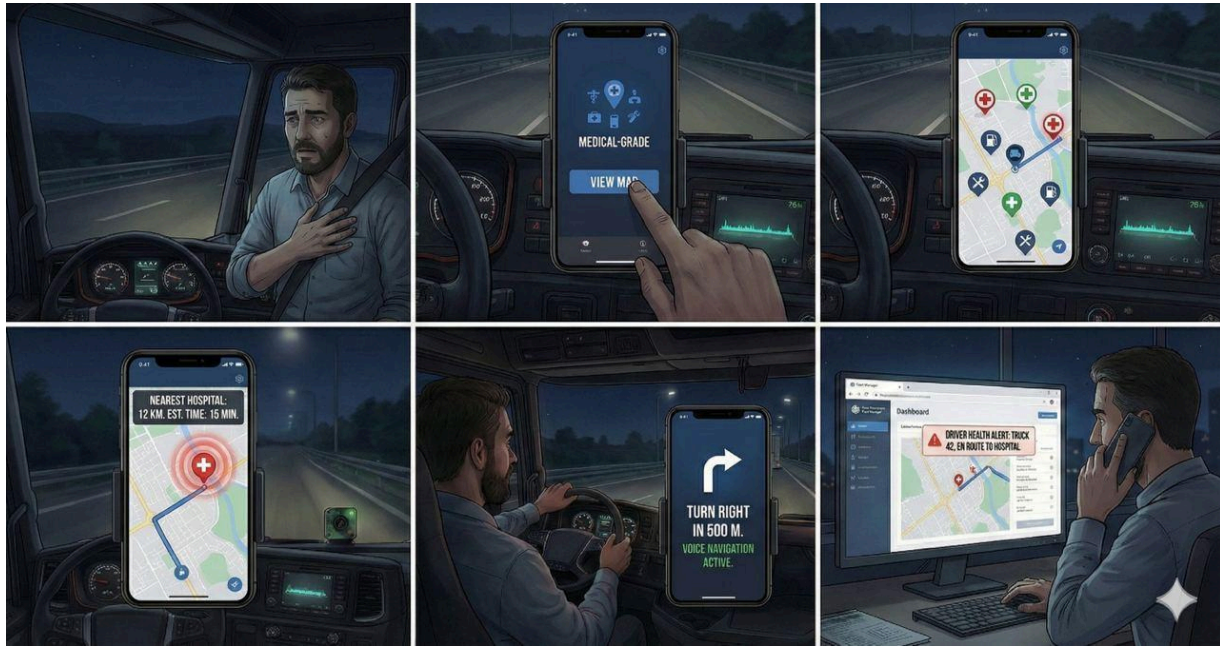


Figure 9: Map-Based Medical and Safety Assistance Navigation for Long-Haul Drivers

Description:

This figure presents a map-based assistance feature designed to support truck drivers in locating nearby medical and safety services during remote highway travel in Egypt. When the driver experiences discomfort while driving, a smartphone securely mounted on the dashboard is used to access the “View Map” function within the mobile application. The system displays a live GPS map showing the truck’s real-time location along with clearly marked nearby hospitals, emergency medical centers, gas stations, and vehicle maintenance facilities. The nearest suitable medical facility is automatically highlighted, with distance, estimated travel time, and a recommended safe route provided. Turn-by-turn voice navigation guides the driver without requiring visual interaction. Simultaneously, the fleet manager monitors the same live map and driver status through the web dashboard. The workflow concludes with the truck arriving safely at the selected facility and the system confirming arrival and safety status, demonstrating effective location-based assistance and coordinated response.

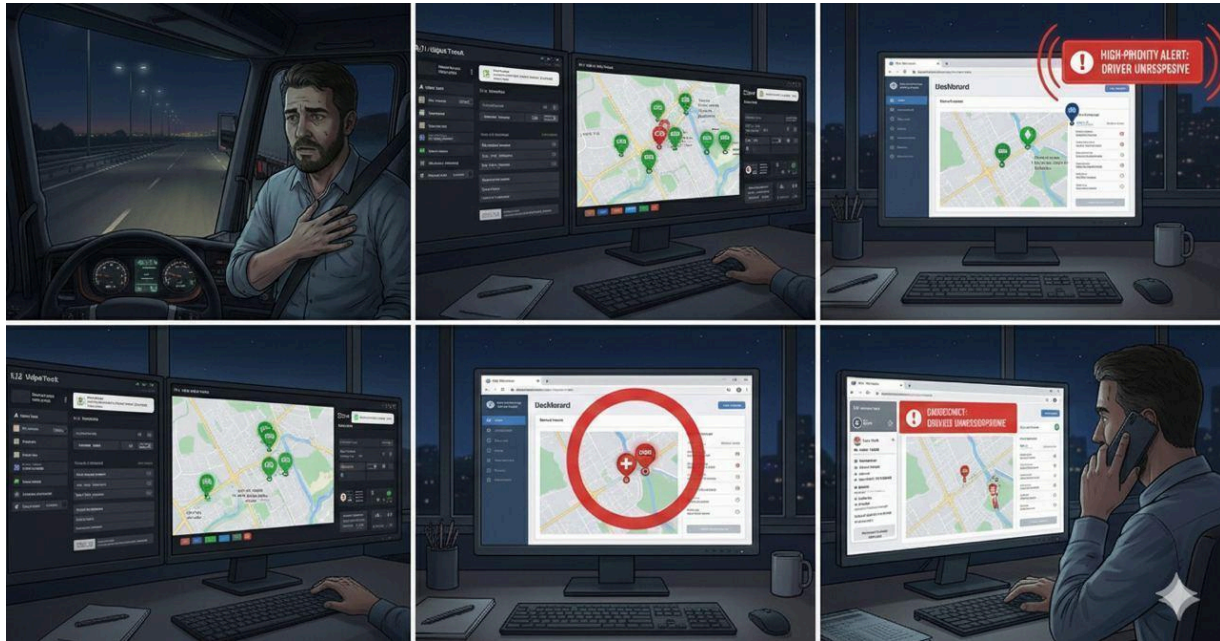


Figure 10: Trip-Based Emergency Escalation and Fleet Manager Notification

Description:

This figure illustrates an automated emergency escalation mechanism based on real-time trip monitoring from the fleet manager's perspective. During an active transport operation, the Udjat Track web dashboard displays multiple trucks moving along their planned routes on a live map. When abnormal behavior is detected for one vehicle, such as prolonged inactivity or sudden loss of driver response, the corresponding truck indicator changes to a high-priority alert state. The fleet manager is immediately notified through the dashboard interface and can access detailed trip information by selecting the alert. The system presents critical data, including the last recorded movement time, an abrupt speed drop to zero, and the exact GPS location of the truck. A clear status message indicates that the driver is unresponsive and requires action, enabling rapid situational awareness and timely decision-making without direct driver input.



Figure 11: Baseline Fleet Trip Monitoring and Safety Awareness

Description:

This figure illustrates normal fleet operations under continuous AI-powered trip monitoring from the fleet manager's perspective. The web-based Udjat Track dashboard presents an overview of multiple active trucks traveling across Egyptian highways, each displayed with a safe operational status. Selecting a vehicle reveals real-time trip information, including route progress, speed, and destination, which updates smoothly without interruption. Driver condition and system indicators remain within normal ranges, with no warnings or escalations triggered. The interface emphasizes clarity and stability, demonstrating how continuous monitoring provides transparency, operational awareness, and confidence during routine transport activities.

3.6 System Modeling

3.6.1 Use Case Diagram

The Use Case Diagram (Figure 12) illustrates the relationships between the actors and the primary system functionalities

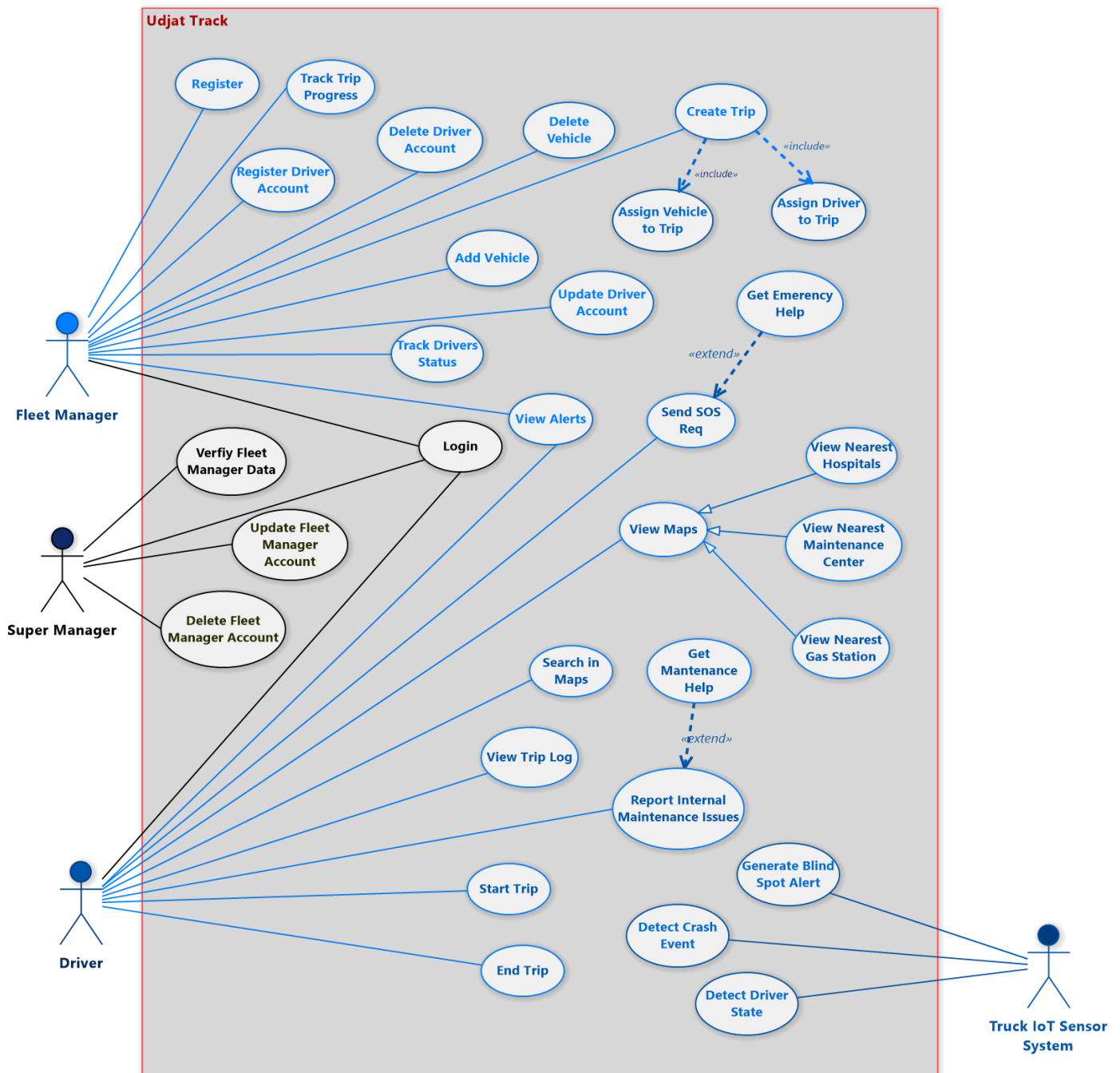


Figure 12: Udjat Track Use Case Diagram

3.6.2 Use Case Tables

Table 4: Report Internal Maintenance Issues Use Case Table

Use Case ID	UC-01	
Use Case Name	Report Internal Maintenance Issues	
Primary Actor	Driver	
Secondary Actor	Maintenance Center	
Included Use Cases	Send Help	
Description	It describes how an internal maintenance issue, such as engine warning, brake issue, abnormal vibration, or system malfunction, is reported by the driver and handled by the system to be delivered to the Maintenance Center, including logging, notification, and follow-up actions by fleet management.	
PreConditions	The driver is authenticated in the mobile application. The vehicle is assigned to the driver.	
Trigger	The driver notices a warning indicating a maintenance issue.	
Main Flow	User Action	System Action
	1- The driver opens the Mobile Application and selects “ Report Maintenance Issue. ”	
		2- The system displays a maintenance reporting form including issue category, severity level, and optional description.
	3- The driver fills in the form’s fields and submits it.	
		4- The system packages the maintenance report payload and transmits it securely to the maintenance center.
Alternative Flows	If the network signal is not stable, the Mobile Application stores the maintenance report locally and will be transmitted automatically once connectivity is restored.	
PostConditions	The maintenance center is notified of the issue and the driver’s current location. The fleet manager is notified. The trip status is updated to Paused or Cancelled , depending on the severity of the maintenance issue.	

Non-Functional Notes	Reliability: Maintenance issue reports shall not be lost due to application crashes or temporary network failures. Availability: The maintenance reporting functionality shall be available 24/7, independent of trip status.
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Table 5: View Alerts Use Case Table

Use Case ID:	UC-02	
Use Case Name:	View Alerts	
Actors:	Driver	
Pre-conditions:	• Driver is authorized and assigned • A trip is active • Mobile application is connected to the Truck IoT Sensor System	
Post-conditions:	• Alert is displayed to the driver. • Alert is logged with timestamp and type. • Driver acknowledgement is recorded.	
Flow of events:	User Action	System Action
	1. Driver is driving during an active trip.	
		2. System continuously monitors safety signals
		3. System detects a condition that requires warning and generates an alert event.
		4. System presents the alert (Visual & Sound).
	5. Driver sees/hears the alert notification.	
	6. Driver acknowledges the alert.	
		7. System records acknowledgement and keeps the alert in the trip log for reporting.
Exception:	User Action	System Action
	1.a.1 Driver continues driving.	
		1.a.2 System detects a general failure in the alert generation module on the

		Raspberry Pi
		1.a.3 System disables real-time alerting functionality and displays: “Alert system temporarily unavailable.”
		1.a.4 System logs the failure event with timestamp and error type for maintenance review.
		1.a.5 System attempts periodic recovery in the background while the driver continues the trip.

Table 6: Add Trip Use Case Table

Use Case ID:	UC-03	
Use Case Name:	Add Trip	
Actors:	Fleet Manager	
Pre-conditions:	1. Fleet Manager is authenticated and logged into the system. 2. Drivers are already registered. 3. Vehicles already added in the system	
Post-conditions:	A new Trip and corresponding assignments are created and stored, and the trip becomes visible to the assigned driver.	
Flow of events:	User Action	System Action
	1. Selects the Create Trip function on the Web Dashboard	
		2. Displays the trip creation form and requests a list of available (idle) drivers and vehicles from the Backend
	3. Enters trip details, including source, destination, and expected start/end times	
		4. Validates the input format and maintains the list of active, verified drivers and vehicles
	5. Selects a specific Vehicle and a qualified Driver to be assigned to the trip	
		6. Checks the AssignmentStatus of the selected entities in the Database to ensure no concurrent active trips exist
	7. Submits the completed trip creation request	

		8. The system validates driver-vehicle compatibility and availability, and a unique ID is created for the trip.
Alternative Flows	Incomplete Data: If the Fleet Manager leaves required fields (like destination) blank, the System prevents submission and highlights the missing fields on the Web Dashboard Network Failure: If the connection between the dashboard and the cloud is lost during submission, the Web Dashboard notifies the User that the trip could not be saved and suggests retrying when connectivity is restored.	
Postconditions	A new Trip and corresponding assignments are created and stored, and the trip becomes visible to the assigned driver.	
Non-Functional Notes	Scalability: The backend must be capable of managing at least 1,000 active vehicles and 100 fleet managers simultaneously without performance degradation during the trip creation process.	

Table 7: Send SOS Alert Use Case Table

Use Case ID	UC-04	
Use Case Name	Send SOS Alert	
Primary Actor	Driver	
Secondary Actor	Fleet Manager, Emergency Services	
Extended Use Cases	Send Emergency Help	
Description	This case describes how the system allows a driver to send an emergency SOS alert either manually or automatically. The alert shares the driver's real-time location, vehicle details, and current driver status with the Fleet Manager and Emergency Services to ensure rapid response during critical situations	
PreConditions	1- Driver is logged into the mobile application 2- Active trip is in progress 3- GPS and network connectivity are available 4- Emergency contact information is configured in the system	
Trigger	1- Driver manually presses the SOS button 2- System detects unconscious driver state 3- IMU sensors detect crash-level impact	
Main Flow	User Action	System Action
	1- Detects an emergency requiring assistance	
	2- Presses the SOS button in the mobile application	
		3- Captures current GPS coordinates and driver status
	4- Displays a confirmation dialog to prevent accidental triggers	
	5- Confirms the SOS request	
		6- Compiles alert package including: • Driver identification • Vehicle identification • Real-time GPS location • Current driver state

		(Alert/Drowsy/High-Risk) • Timestamp • Trip details
		7- Transmits alert to Fleet Manager's dashboard
		8- Transmits alert to Emergency Services
		9- Displays confirmation message to driver
		10- Continues to share live location updates until help arrives
Alternative Flows	1- IMU sensors detect crash-level impact 2- System initiates countdown timer for driver responsiveness 3- System displays visual alert and plays audio warning 4a-. If driver responds: Cancel automatic SOS, log crash event 4b- If driver does not respond: • System automatically triggers SOS alert • Mark alert as "Crash-related".	
PostConditions	1-SOS alert is successfully transmitted to the Fleet Manager and Emergency Services 2- The driver's real-time location and status are shared 3- Alert is logged in the system for audit purposes 4- Fleet Manager receives notification on the Web Dashboard	
Non-Functional Notes	This use case implements FR-18 (Manual SOS) and FR-19 (Automatic SOS) SOS alerts must be transmitted with minimal latency Location and alert data must be securely transmitted and logged.	

Table 8: Generate Blind-Spot Alert Use Case Table

Use Case ID:	UC-05	
Use Case Name:	Generate Blind-Spot Alert	
Actors:	Truck IoT Sensor System	
Pre-conditions:	• Ultrasonic sensors are powered and initialized. • Communication link to the Raspberry Pi/mobile app is available	
Post-conditions:	• A blind-spot alert event is generated and sent to the local application. • The alert event is logged locally.	
Flow of events:	Actor Action	System Action
	1. Samples ultrasonic distance readings at a fixed interval.	
	2. Provides distance measurement (cm) for each blind-spot sensor.	
		3. Compares distance values to the blind-spot threshold.
		4. If the distance is below the threshold, it creates a Blind-Spot Alert event.
		5. Displays an alert to the driver (sound/visual)
Exception:	Actor Action	System Action
	1.a.1 Sensor fails self-test or stops responding.	
		1.a.2 System marks blind-spot monitoring as unavailable and shows: "Blind-spot monitoring unavailable."
		1.a.3 Logs failure and continues monitoring other available sensors

Table 9: Detect Crash Event Use Case Table

Use Case ID:	UC-06	
Use Case Name:	Detect Crash Event	
Actors:	Truck IoT Sensor System	
Pre-conditions:	- IMU is powered, calibrated, and sampling. - Local system clock is available for event timestamping.	
Post-conditions:	- Crash event is generated. - Crash event is delivered to the local alert module and stored for backend sync.	
Flow of events:	Actor Action	System Action
	1. Continuously samples IMU acceleration and gyroscope values.	
	2. Provides updated IMU reading stream.	
		3. gyroscope values. Receives IMU readings and computes magnitude features.
		4. Compares computed magnitudes against crash thresholds.
		5. If thresholds are exceeded, it confirms the crash condition and creates a Crash Event.
		6. Triggers emergency workflow and logs the event.
Exception:	Actor Action	System Action
	1.a.1 IMU stops responding or fails self-test.	
		1.a.2 System disables crash detection and displays: "Crash detection unavailable."
		1.a.3 Logs failure and attempts background reinitialization.

Table 10: Detect Driver State Use Case Table

Use Case ID:	UC-7	
Use Case Name:	Detect Driver State	
Actors:	Truck IoT Sensor System	
Pre-conditions:	The in-cabin camera feed is available and functioning properly.	
Post-conditions:	- Driver-state event is generated. - Alert is generated if the state is risky and logged for reporting.	
Flow of events:	Actor Action	System Action
	1. Captures driver-state input stream	
	2. Provides continuous driver-state signals.	
		3. If a risky state is detected, it generates a Driver State Alert.
		4. If thresholds are exceeded, it confirms the crash condition and creates a Crash Event.
		5. Notifies driver (sound/visual) and logs the event.
Exception:	Actor Action	System Action
	1.a.1 Detection component fails unexpectedly.	
		1.a.2 System restarts the module in the background, logs the error, and temporarily disables driver-state alerts.

3.6.3 Class Diagram

3.6.3.1 Backend Class Diagram

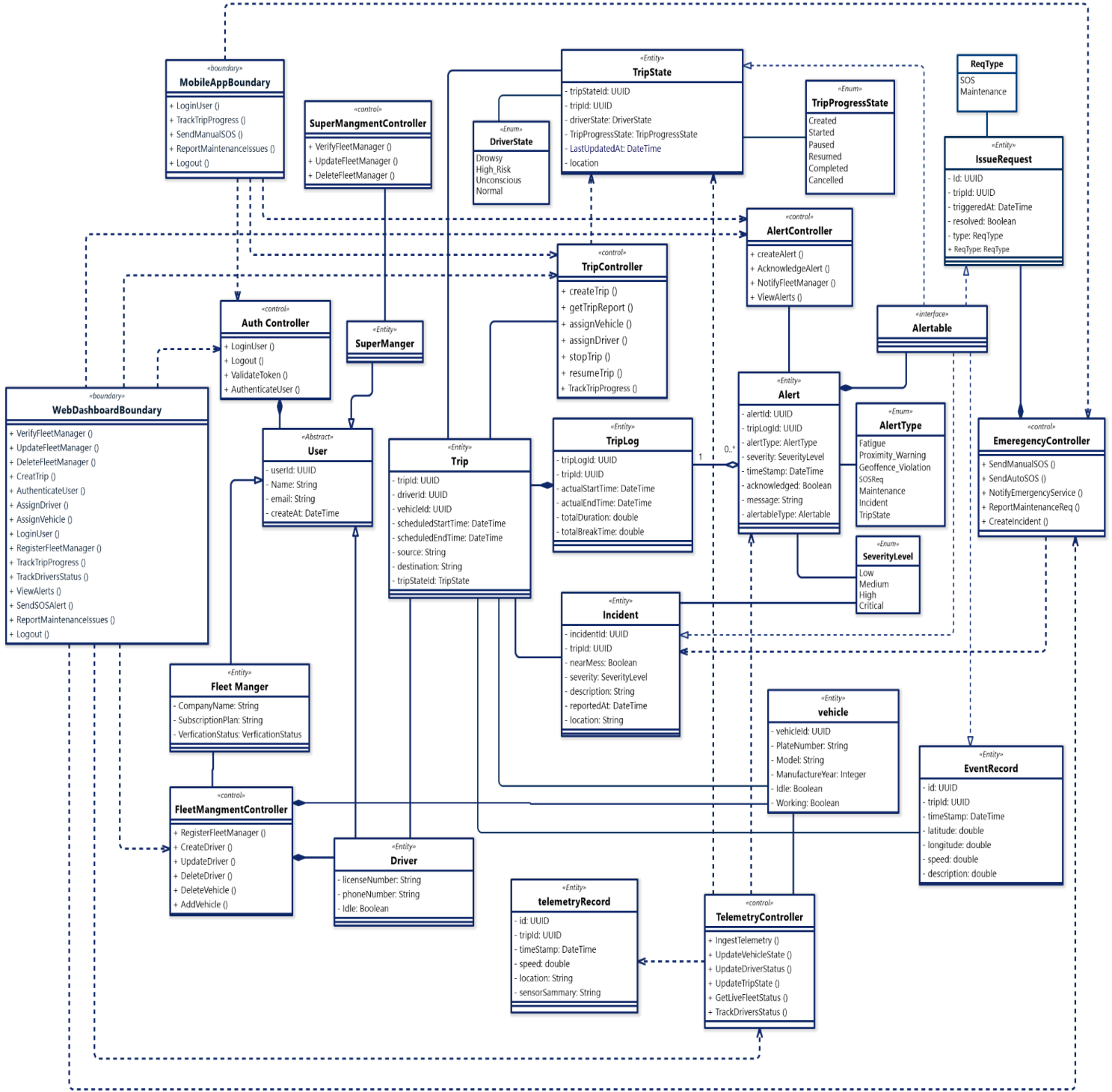


Figure 13: Udjat Track Backend Class Diagram.

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3.6.3.3 Mobile Class Diagram

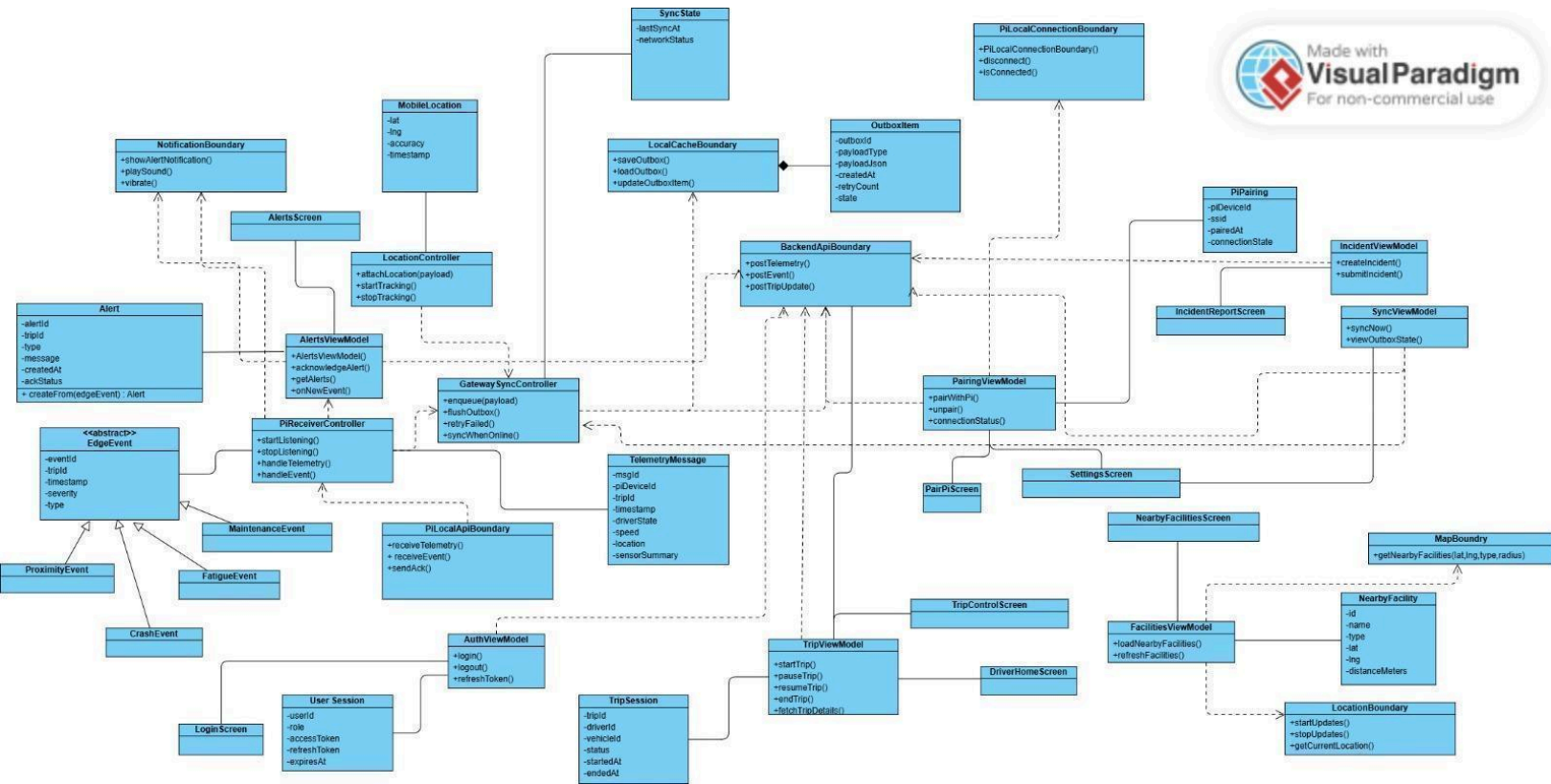


Figure 15: Udjat Track Mobile Class Diagram.

The Entity Relationship Diagram (Figure 16) defines the logical structure and associations of the persistent data stored within the cloud database system



3.6.5 IoT Sequence Diagram

3.6.5.1 Crash Detection Sequence Diagram

The Crash Detection Sequence Diagram (Figure 17) depicts the process of detecting crash-level impacts from IMU sensor data and generating a corresponding crash event.

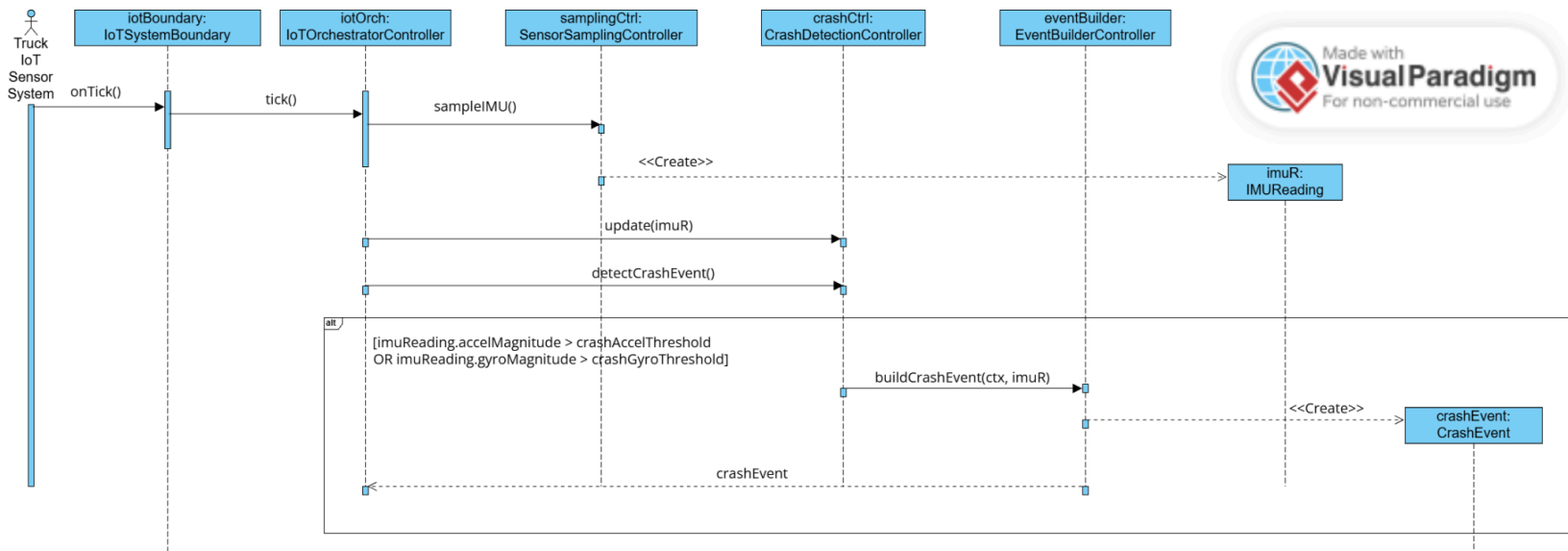


Figure 17: The Crash Detection Sequence Diagram.

3.6.5.2 Periodic Sensor Sampling Sequence Diagram

The Periodic Sensor Sampling Sequence Diagram (Figure 18) shows how the system periodically collects data from ultrasonic and IMU sensors and generates structured sensor readings for further processing.

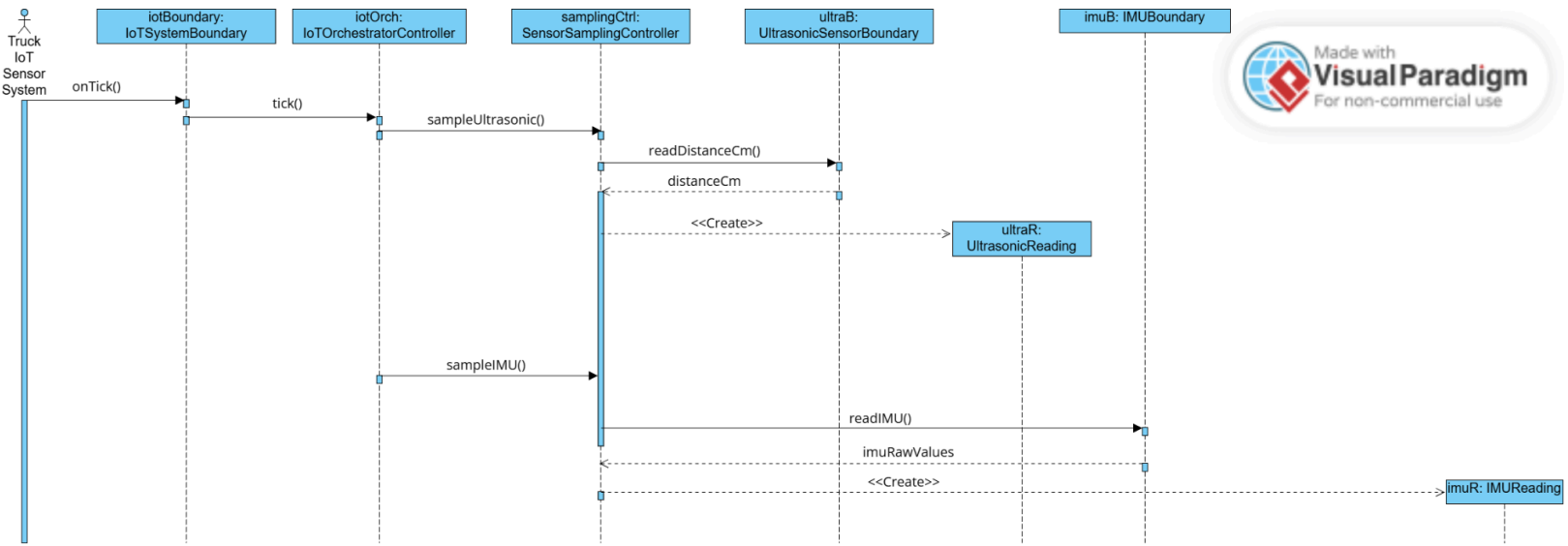


Figure 18: Periodic Sensor Sampling Sequence Diagram

3.6.5.3 Proximity Event Detection Sequence Diagram

The Proximity Event Detection Sequence Diagram (Figure 19) illustrates how ultrasonic distance readings are evaluated against proximity thresholds to generate proximity warning events.

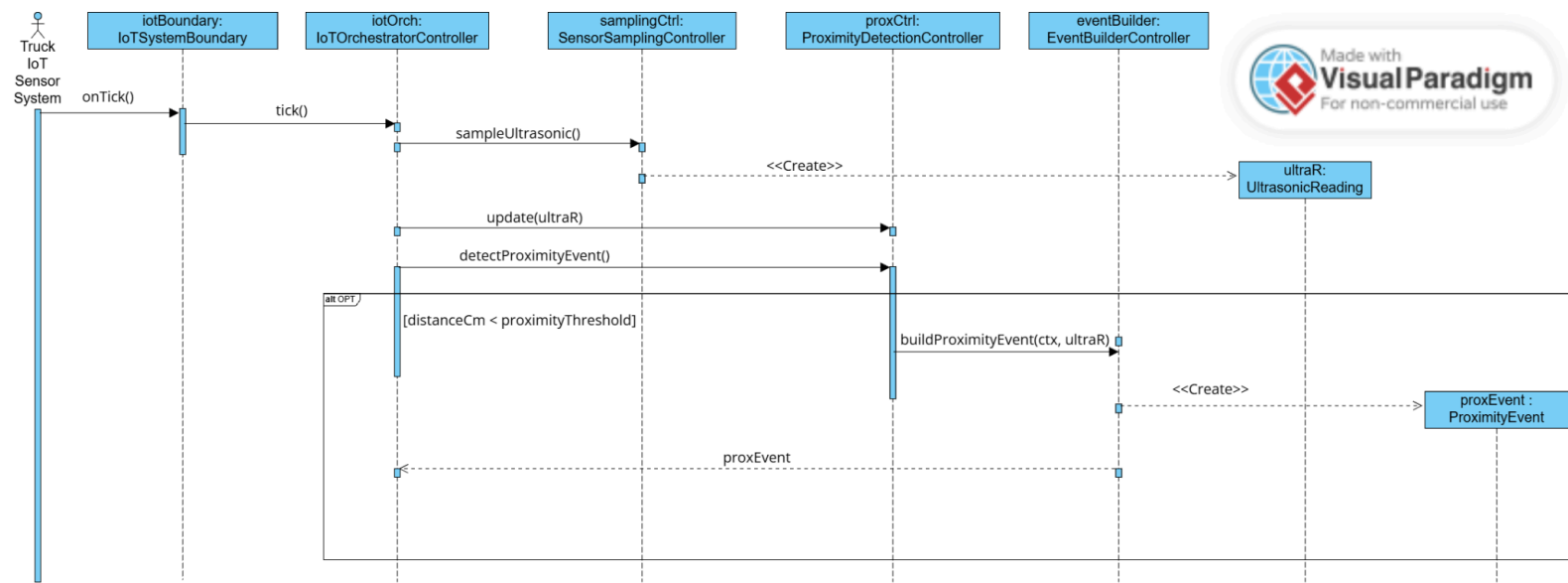


Figure 19: Proximity Event Detection Sequence Diagram

3.6.6 Mobile Sequence Diagram

3.6.6.1 Driver Alert Generation and Notification Sequence Diagram

The Driver Alert Generation and Notification Sequence Diagram (Figure 20) shows how detected events are transformed into driver alerts and safely queued for backend synchronization.

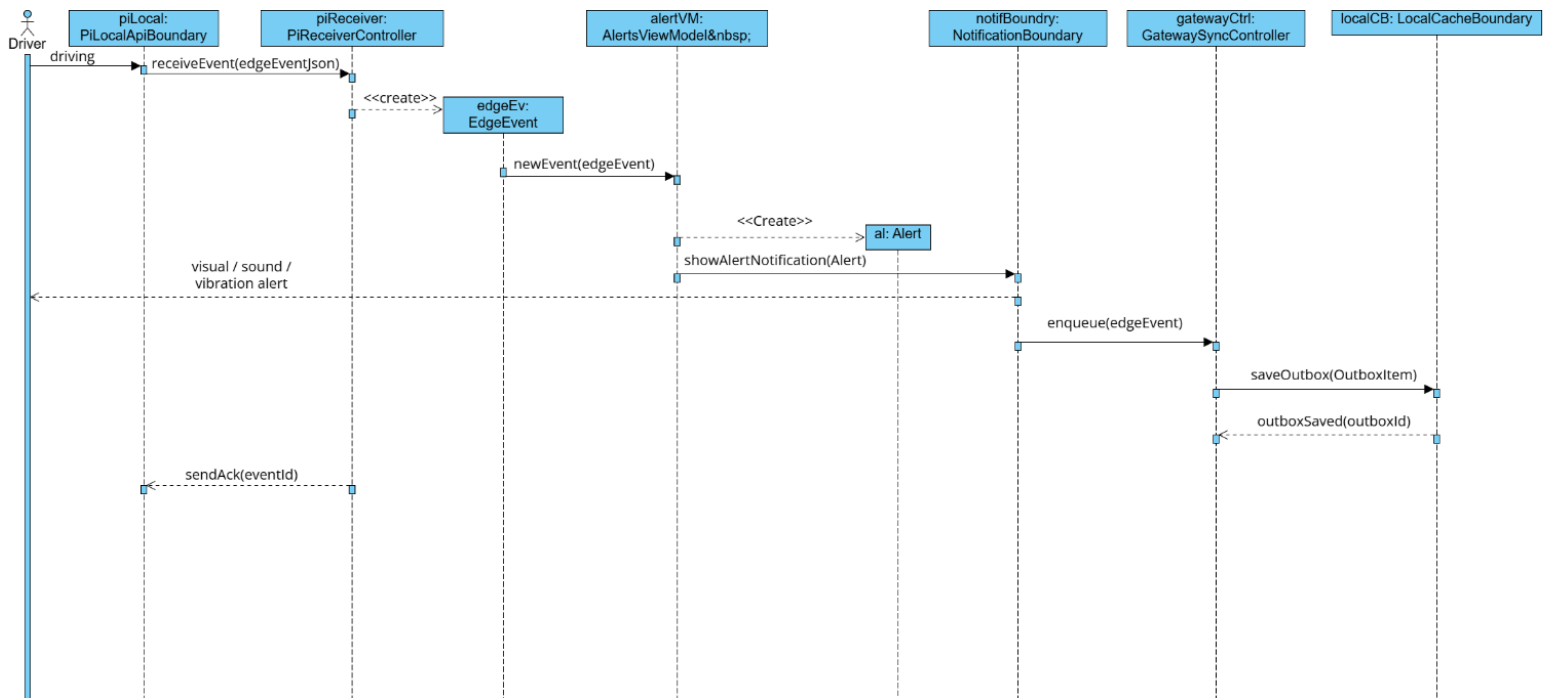


Figure 20: Driver Alert Generation and Notification Sequence Diagram

3.6.6.2 Nearby Facilities Retrieval Sequence Diagram

The Nearby Facilities Retrieval Sequence Diagram (Figure 21) illustrates how the system obtains the driver's current location and retrieves a list of nearby facilities, which is then displayed on the map interface.

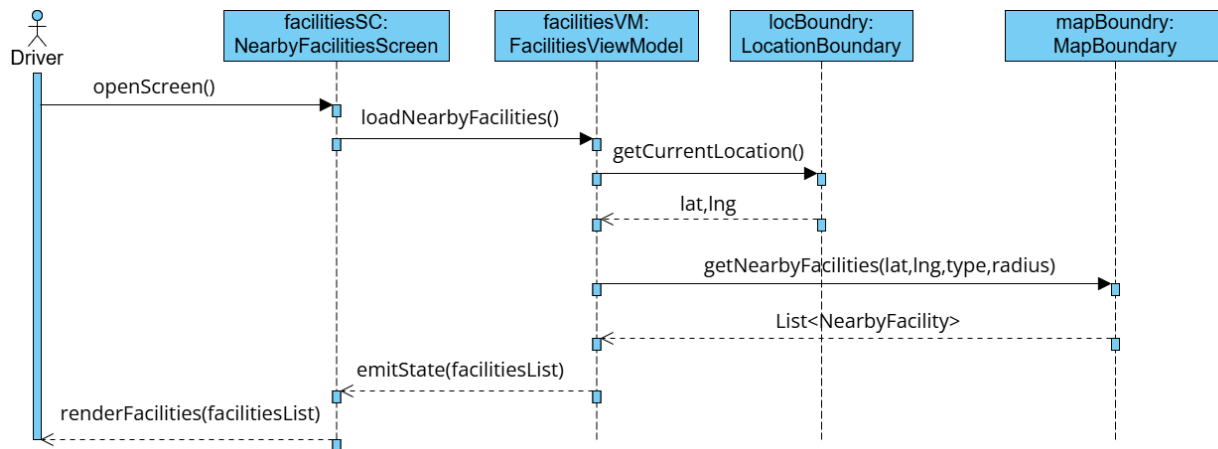


Figure 21: Nearby Facilities Retrieval Sequence Diagram

3.6.6.3 Offline-to-Online Data Synchronization Sequence Diagram

The Offline-to-Online Data Synchronization Sequence Diagram (Figure 22) shows how locally stored data is reliably sent to the backend once network connectivity is restored.

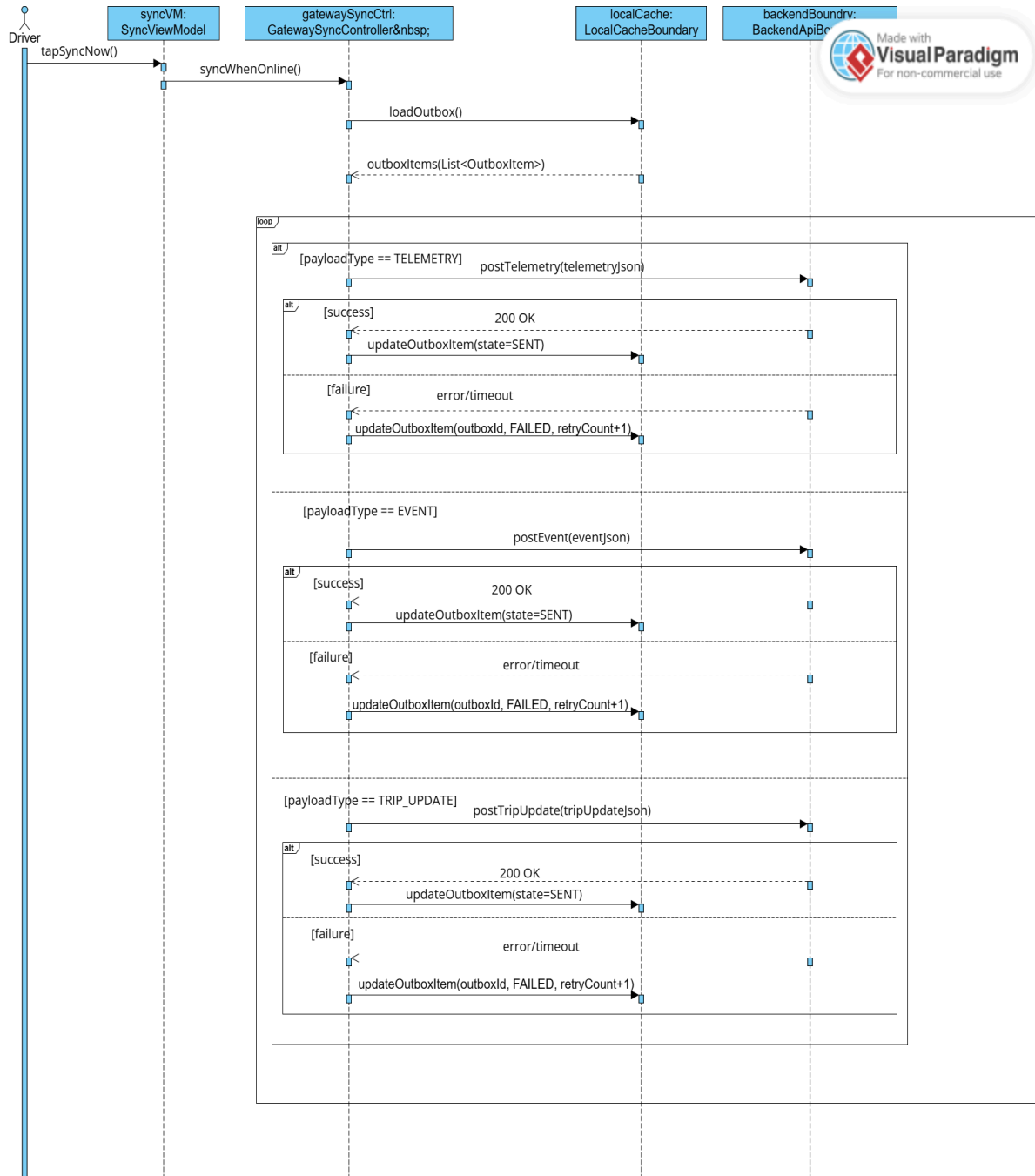


Figure 22: Offline-to-Online Data Synchronization Sequence Diagram

3.6.7 Backend Sequence Diagram

3.6.7.1 Delete Vehicle Sequence Diagram

This delete Vehicle Sequence Diagram (Figure 23) illustrates the administrative process used by the Fleet Manager to remove a vehicle from the company's active fleet.

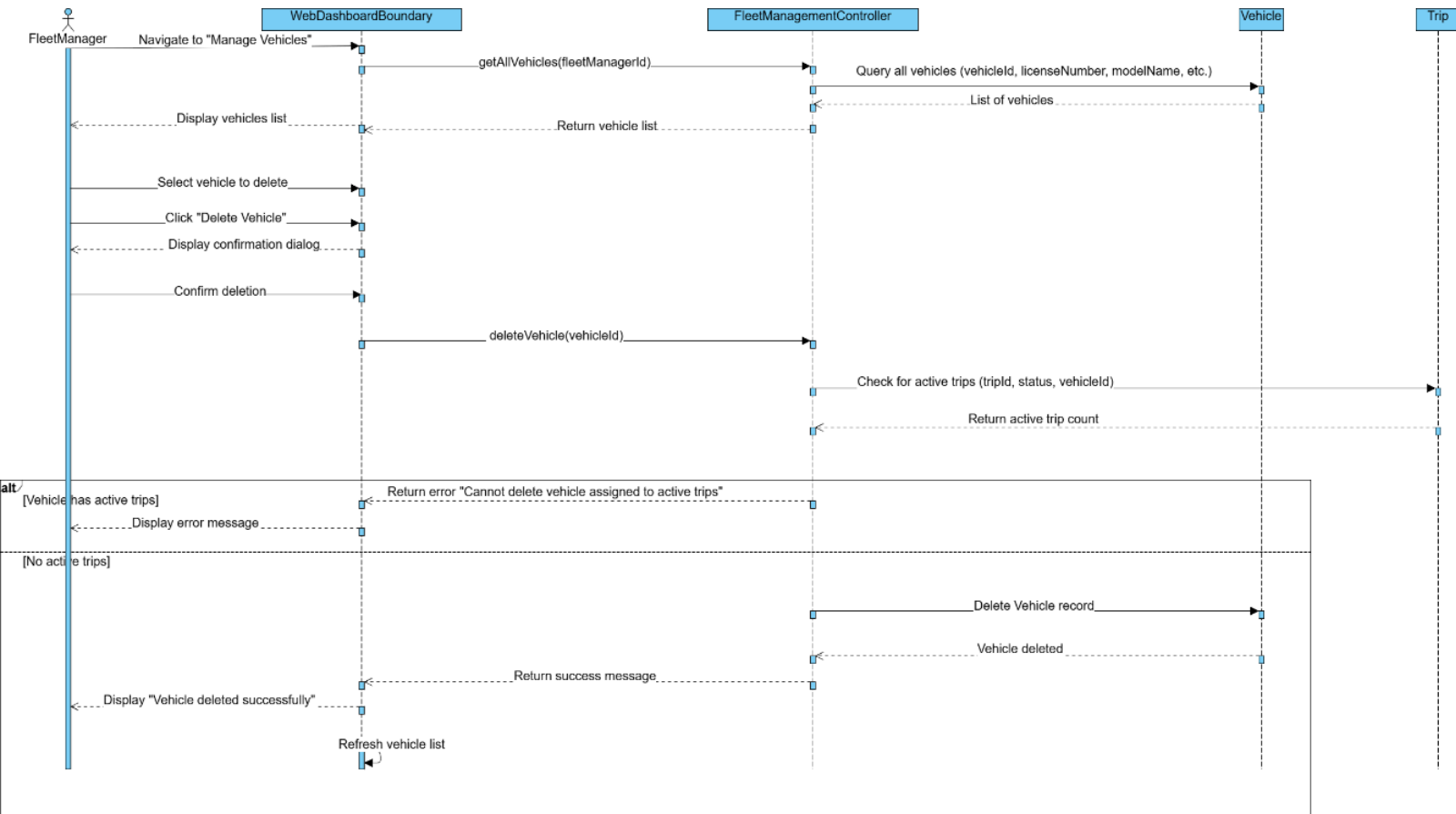


Figure 23: Delete Vehicle Sequence Diagram

This track Driver Status Sequence Diagram (Figure 24) describes the real-time monitoring of a driver's physiological and operational state by the **Fleet Manager**.



Chapter 6 Conclusions and Future Work

6.1 Conclusions

The primary objective of the Udjat Track project is to address the persistently high rate of road traffic accidents in Egypt, which account for approximately 12,000 fatalities each year. Heavy transport vehicles represent a significant proportion of this burden, contributing to nearly 27% of recorded crashes and exhibiting a substantially higher accident involvement rate than passenger vehicles. In response to this risk profile, the project focuses on developing a targeted safety intervention for heavy transport operations. The proposed system is designed to tackle key contributing factors including driver fatigue, distraction, and delayed emergency response through a layered Edge–Gateway–Cloud architecture intended to support real-time detection, timely alerts, and reliable post-incident communication.

The ongoing implementation of Udjat Track is structured around several core functionalities aimed at mitigating accident risk:

Preventive AI Monitoring: The system is being developed to utilize on-device computer vision for real-time detection of drowsiness and micro-sleeps, enabling early warnings before hazardous situations escalate.

IoT Safety Retrofitting: Ultrasonic sensors are being integrated to support blind-spot and forward-collision warnings, providing a low-cost safety enhancement for older Egyptian truck models that typically lack Advanced Driver Assistance Systems (ADAS).

Emergency Response Optimization: An automated SOS mechanism is being implemented based on IMU-driven crash detection and driver consciousness assessment, with the goal of transmitting live GPS coordinates and driver status to emergency services without delay.

Fleet Accountability: A web-based dashboard is being developed to enable fleet managers to transition from passive tracking to active safety monitoring by providing real-time oversight and historical trip and safety logs for compliance and insurance review.

Overall, the Udjat Track project demonstrates the potential of combining edge processing with modular IoT hardware to deliver an affordable, proactive, and scalable safety solution tailored to the needs of the Egyptian heavy transport sector. While the system is still under implementation, its design and preliminary development indicate a promising approach toward improving road safety and reducing accident-related risks in logistics operations.

6.2 Future Work

As Udjat Track is still under development, several directions for future enhancement are identified to further strengthen its functionality, scalability, and practical deployment in real-world heavy transport operations:

Advanced AI and Environmental Adaptation: Future development should focus on improving AI inference performance under challenging environmental conditions such as direct sunlight, night-time driving, or low-visibility scenarios. Expanding the training dataset to include a more diverse range of driver demographics, facial features, and driving environments would also enhance model generalization and robustness across different operational contexts.

Enhanced Predictive Maintenance: Further research could explore the use of advanced machine learning techniques on long-term IMU and vehicle telemetry data to estimate the remaining useful life of critical vehicle components. Such predictive maintenance capabilities have the potential to reduce unexpected breakdowns and optimize maintenance scheduling for fleet operators.

Insurance Integration: A dedicated API could be developed in future iterations to enable integration with insurance providers. This would allow for dynamic “pay-how-you-drive” or usage-based insurance models based on safety scores generated by the system, creating financial incentives for safer driving behavior.

Intelligent Automated Break Management System: A future enhancement could introduce an intelligent break scheduling module that analyzes driving duration, fatigue indicators, and route characteristics to recommend optimal rest locations along the journey. The system could identify nearby rest stops or safe parking zones and proactively notify drivers when breaks are advisable. Additionally, a mandatory break enforcement mechanism could be considered, temporarily restricting trip continuation in cases of excessive fatigue or when regulatory driving-hour limits are approached.

Steering Wheel Grip Detection and Driver Engagement Monitoring: Subsequent versions of the system could integrate steering wheel grip sensors or vision-based hand detection to continuously assess driver engagement. If prolonged disengagement or lack of response to steering prompts is detected, the system could initiate escalating alerts and, in critical cases, trigger emergency protocols such as automatic SOS activation or vehicle deceleration. This would be particularly valuable for detecting medical emergencies or severe fatigue episodes.

Intelligent Trip Scheduling and Route Optimization: Future work may include the development of an advanced trip planning module that assists fleet managers in optimizing schedules based on driver work-hour regulations, predicted fatigue patterns, maintenance requirements, traffic conditions, and weather forecasts. The system could dynamically adjust routes and break recommendations during active trips to ensure both efficiency and regulatory compliance.

Flexible Subscription and Pricing Models: To support broader adoption, future deployments could introduce tiered subscription models tailored to different fleet sizes, vehicle types, and functional requirements. Pricing could be based on the number of active vehicles, data storage needs, and level of technical support required, ensuring accessibility for both small operators and large enterprise fleets.

Overall, these proposed enhancements aim to evolve Udjat Track into a more intelligent, adaptive, and commercially viable safety platform for heavy transport operations in Egypt and beyond.

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